DAVI CARVALHO ARAÚJO

Product Development Coordinator | Combustion & Aftertreatment Control Systems | Model-Based Design | AUTOSAR | Hybrid & Electric Propulsion

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PROFILE

Mechanical Engineer specialized in control strategy development for propulsion systems (ICE, hybrid, and electric), currently leading the Combustion & Aftertreatment control team at Stellantis as Product Development Coordinator. Experienced across the full model-based development lifecycle — from functional requirements to control design, implementation, and MIL/SIL/HIL validation — on engine platforms including the Hurricane 4 EVO, Hurricane Twin Turbo I-6, and Pentastar 3.6 V6. Recently completed a specialization in Hybrid and Electric Vehicles and Vehicle Dynamics & Chassis.

EXPERIENCE

Product Development Coordinator | Stellantis – São Paulo, SP *Sep 2024 – Present*

- **Technical ownership of engine control portfolio:** Product Owner for Combustion & Aftertreatment control functions across the Hurricane 4 EVO (inline-4 turbo with Turbulent Jet Injection and multiple-pulse PDI, and GPF-equipped variant for particulate matter control), the Hurricane Twin Turbo I-6 (inline-6 bi-turbo, standard and high-output), and the Pentastar 3.6 V6 (V6 PFI with dual-stage valve lifting, integrated into a REEV hybrid architecture).
- **Requirements and control design:** Lead the definition of high-level functional requirements and their translation into physics-based control logic — spanning spark timing, knock control, fuel dynamics, thermal management, and exhaust aftertreatment (GPF, oxygen sensors, air-fuel ratio control) — ensuring traceability between requirements, design, and test cases.
- **Design approval and release management:** Review and approve control designs, manage production release content, and prioritize the backlog for software documentation, requirements review, and validation test cases.
- **Technical leadership and issue resolution:** Drive cross-functional technical discussions to diagnose and resolve complex combustion and aftertreatment system issues, acting as the primary point of contact for implementation across teams.
- **Supplier and cross-team integration:** Interface directly with BSW suppliers and system architecture teams (AUTOSAR) to align connected designs that meet functional and safety requirements.
- **Agile execution:** Operate in a fast-paced agile environment, prioritizing deliverables against vehicle program launch timelines.

Product Engineer | Bosch – Campinas, SP *Jan 2023 – Sep 2024*

- **Control strategy development:** Designed and implemented control strategies for conventional and hybrid propulsion systems across all Stellantis engine platforms, covering spark timing, knock detection, air-fuel ratio control, and exhaust thermal management.
- **Model-based software development:** Built and maintained physics-based control models in MATLAB/Simulink, translating functional requirements into simulation-ready logic for combustion and aftertreatment systems.
- **System architecture:** Defined AUTOSAR software architecture in dSpace SystemDesk, structuring software components and interfaces for integration across the vehicle's control system.
- **Combustion and aftertreatment design:** Developed new control design solutions for combustion strategies (e.g., spark and knock control) and aftertreatment functions (e.g., GPF regeneration, oxygen sensor management), from concept through implementation.
- **Issue resolution:** Investigated and resolved software and calibration issues identified during vehicle testing and post-production, tracing root causes to control logic or requirements gaps.
- **Verification and validation:** Planned and executed software testing using MIL, SIL, and HIL methodologies to validate control logic against functional requirements before release.
- **Embedded integration:** Integrated C-code control logic onto embedded hardware targets, supporting the transition from model-based design to production software.
- **Agile delivery:** Worked within an agile/autoscrum framework, serving as Scrum Master to coordinate sprint planning and remove delivery blockers.
- **Global collaboration:** Partnered with a global cross-functional team at Stellantis to align control strategies across regions and engine programs.

Young Talent Internship Program | Bosch – Campinas, SP *Sep 2022 – Jan 2023*

- **Propulsion systems fundamentals:** Built foundational expertise in ICE, hybrid, and electric vehicle propulsion systems, including their control and calibration principles.
- **Engine and transmission calibration:** Supported calibration activities for ICE systems across automatic (AT), manual (MT), and continuously variable (CVT) transmissions.
- **Requirements engineering:** Contributed to the definition of high-level functional requirements for new feature development.
- **Model-based simulation:** Applied MATLAB/Simulink for Model-in-the-Loop (MIL) simulation and validation of control logic.
- **Software development:** Developed scripts and tools in Python and C++ to support calibration and data analysis workflows.
- **Calibration data management:** Managed calibration datasets using ETAS INCA, supporting traceability across development iterations.

EDUCATION

Postgraduate Specialization | Instituto Mauá de Tecnologia *2025 – Present*

- Hybrid and Electric Vehicles
- Vehicle Dynamics and Chassis

BS in Mechanical Engineering | Universidade Federal de Sergipe *2018 – 2023*

- Grade: 8.5/10.0
- Coursework: Fuel Heating Control System Development using Model-Based Design methodology

TECHNICAL SKILLS

- **Software & Tools:** MATLAB/Simulink, AUTOSAR, ETAS INCA/MDA, CANoe, CANalyzer;
- **Programming Languages:** C, C++, Python;
- **Methodologies:** Model-Based Design, MIL/SIL/HIL Testing, Agile/Scrum;
- **Domains:** Combustion Control (Cylinder Fuel Mass Targeting, Lambda Control, Closed-Loop Fueling, Spark Timing Control, Knock Detection and Mitigation, Cylinder-Level Coordination); Aftertreatment Systems (Catalyst Temperature Management, Exhaust Oxygen Sensing, GPF, Exhaust Temperature/Pressure Protection, Active Exhaust Valve Control).